LLM Model Selection Report

Egyptian Arabic conversational models — bake-off results

Best robot realtime	Best quality	Lowest TTFT	OK / attempted
Nile-Chat-4B	**Qwen3-8B**	**Qwen3-4B-Instruct-2507**	**4 / 7**

Overview

This report compares 7 LLM candidates for an Egyptian Arabic voice robot.
4 models completed the suite successfully; 3 failed (typically gated Hugging Face auth).
Two rankings matter: (1) robot realtime = latency + throughput + VRAM + load, and
(2) response quality = rubric scores for correctness, instruction following, and TTS suitability.
Fastest composite pick: Nile-Chat-4B.
Highest quality pick: Qwen3-8B (4.65/5).
These two leaders may differ — choose based on whether turn-taking speed or answer quality is the priority.

How to read this report

Page 2: metrics glossary — what every term means and whether LOWER or HIGHER is better.
Page 3: recommended picks with rationale.
Pages 4-6: latency, throughput, VRAM, and normalized score charts.
Pages 7-9: quality dimensions, category heatmaps, and score distributions.
Page 10: decision guidance for production selection.

LLM - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

TTFT - Time To First Token (s)

LOWER better

Delay until the model starts generating. Lower = snappier turn-taking (robot starts sooner).

Throughput (tokens/s)

HIGHER better

How fast tokens are generated after the first one. Higher = long answers finish sooner.

Generate time (s)

LOWER better

Total time to finish the reply. Lower is better (also depends on answer length).

Quality / overall score (/5)

HIGHER better

Rubric score for answer quality. Higher = better answers for the robot.

Auto-pass rate %

HIGHER better

Share of prompts that passed automated checks. Higher = more reliable across the suite.

Correctness / Instruction / Conciseness / TTS fit (/5)

HIGHER better

Quality sub-scores. Higher on each is better. Conciseness + TTS fit matter for spoken replies.

Latency / Throughput / VRAM / Load scores (0-100)

HIGHER better

Normalized component scores inside the robot composite. Higher = better on that axis.

Peak RAM (MB)

LOWER better

System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better

CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better

Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better

Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

x Realtime (xRT)

HIGHER better

How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Robot realtime score (0-100)

HIGHER better

Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Balanced score (0-100)

HIGHER better

Normalized blend of the main tradeoffs. Higher = better all-rounder.

Remember

For LLM robots: want LOWER TTFT, LOWER VRAM/RAM, LOWER load time;
want HIGHER tokens/s, HIGHER quality scores, HIGHER auto-pass %, HIGHER robot score.
Prefer quality when answers are spoken aloud; prefer TTFT when turn-taking feel matters most.

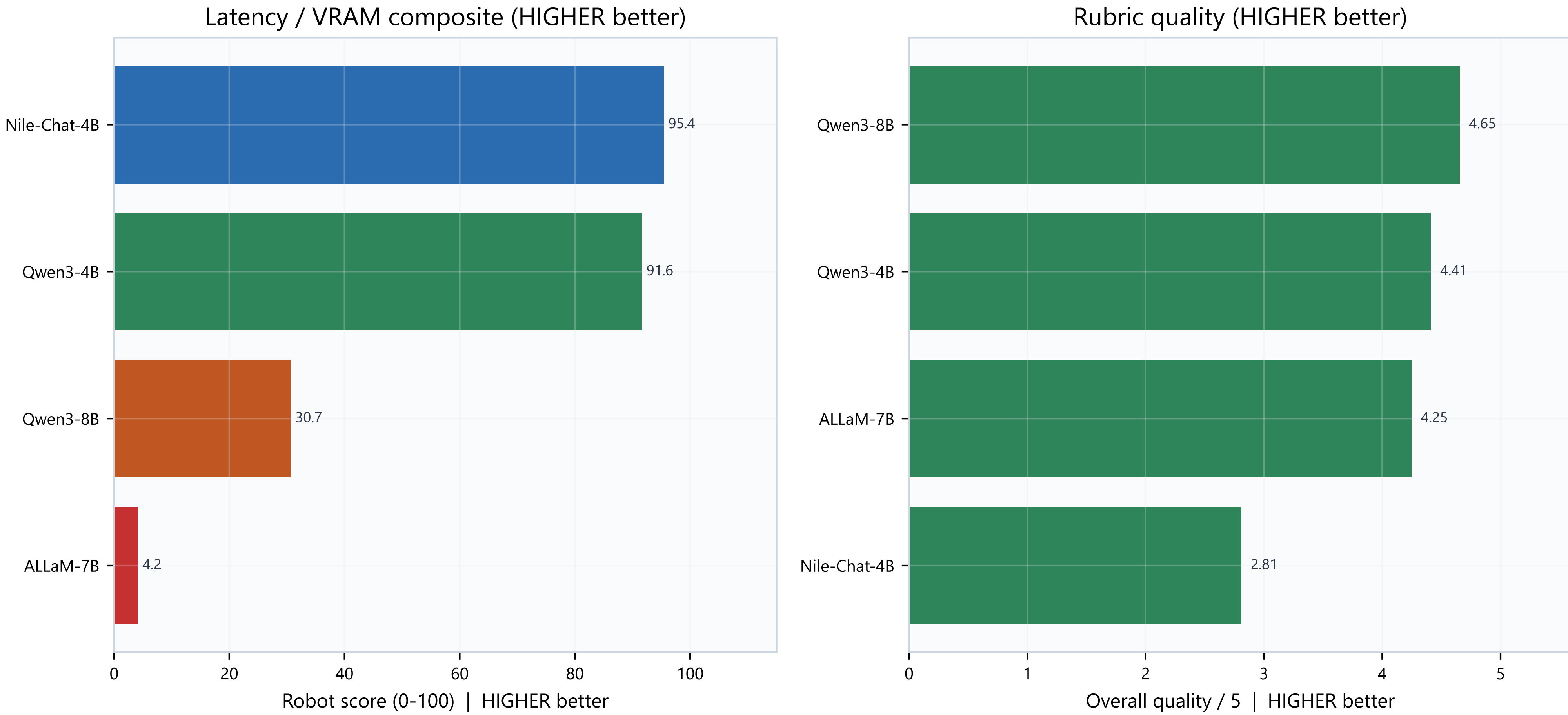
Recommended picks (automated)

Best for robot realtime **Nile-Chat-4B**	Best composite of TTFT + tok/s + VRAM + load for conversational turns. score_robot_realtime=95.41 | avg_first_token_seconds=0.96 | avg_tokens_per_second=11.56 | peak_vram_mb=8394.2	
Lowest time-to-first-token (LOWER better) **Qwen3-4B-Instruct-2507**	Fastest average time-to-first-token (best turn-taking feel). avg_first_token_seconds=0.947	
Highest throughput (HIGHER better) **Nile-Chat-4B**	Highest average tokens/second. avg_tokens_per_second=11.56	
Lowest peak VRAM (LOWER better) **Qwen3-8B**	Lowest peak VRAM — better for small GPUs / co-residency. peak_vram_mb=6900.2	
Best balanced **Nile-Chat-4B**	Balanced latency/throughput/VRAM tradeoff. score_balanced=93.75	

Selection notes

Automated scores cover latency, throughput, and resources only — not dialect naturalness.
Manually spot-check Egyptian / MSA / code-switch responses under llm_outputs/responses/.
For robot UX, prioritize LOW TTFT (lower better) even if peak tok/s is slightly lower.
Disable thinking modes where available to reduce TTFT further.

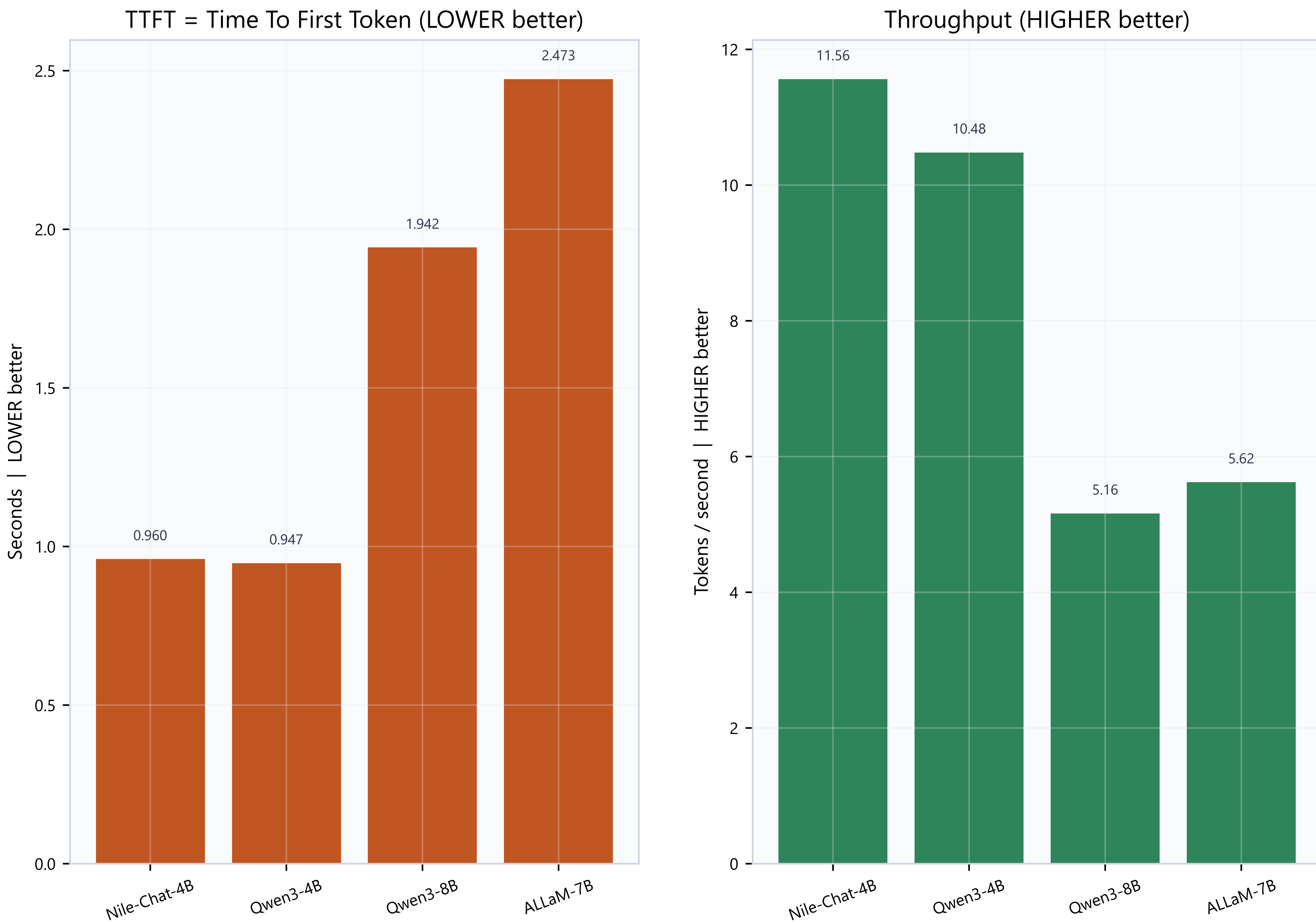
Robot realtime score vs Response quality



What this means

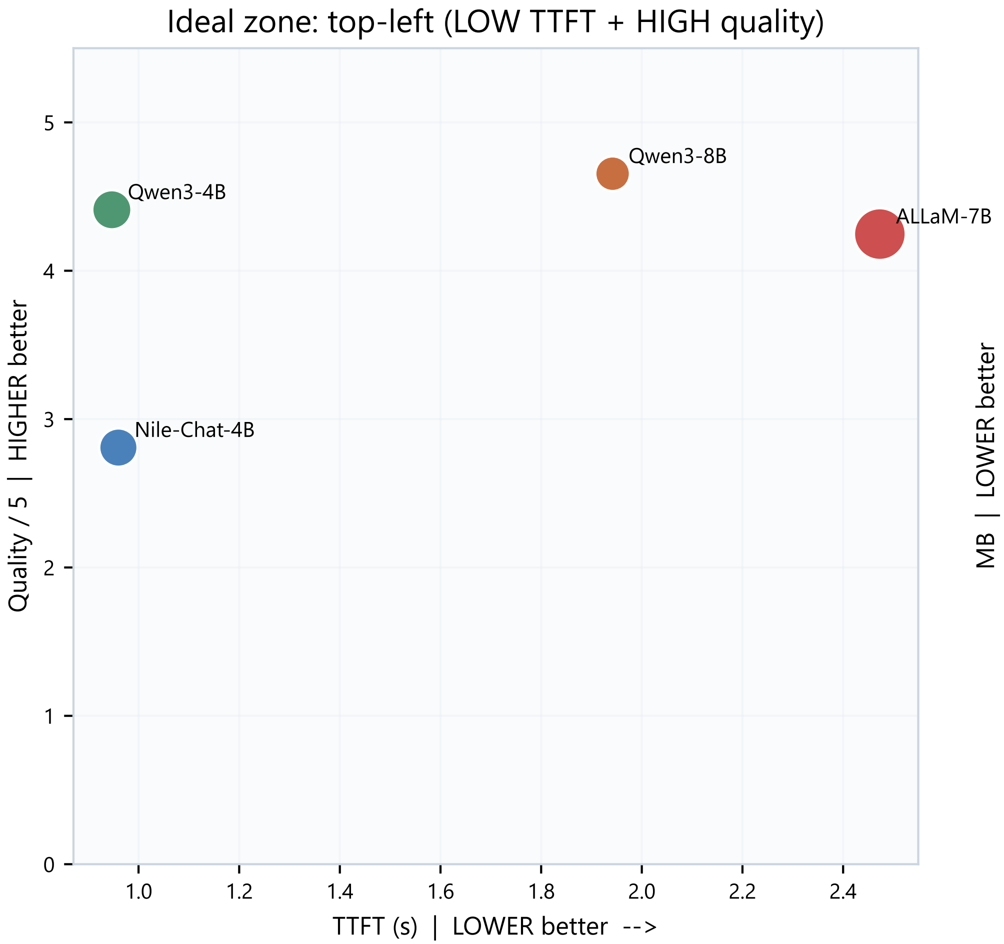
Robot score (HIGHER better): blends TTFT, tokens/s, VRAM efficiency, and load time.
Quality /5 (HIGHER better): rubric for correctness, instructions, conciseness, TTS fit.
Left chart: Nile-Chat-4B leads robot realtime.
Right chart: Qwen3-8B leads at 4.65/5.
Nile-Chat-4B quality is 2.81/5 — strong on speed, weaker on rubric quality.
Prefer left leader for snappy turns; prefer right leader for better spoken answers.

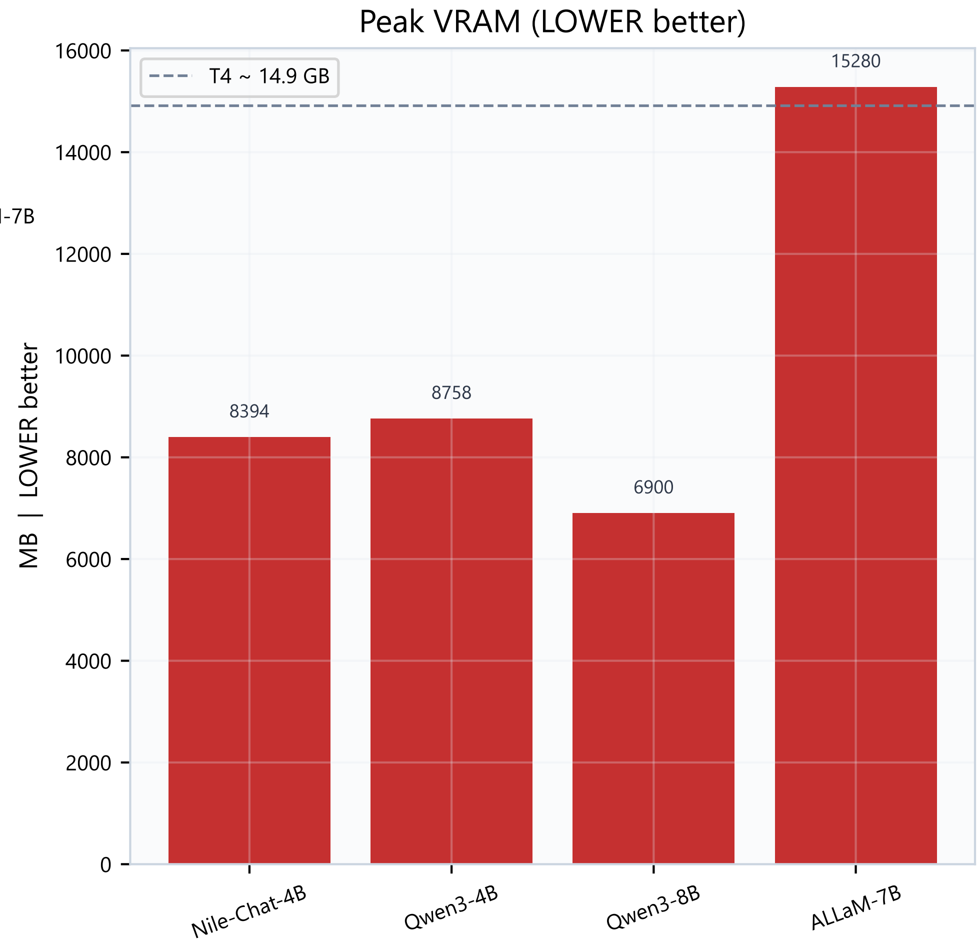
Latency & throughput



TTFT (LOWER better): delay before the first token — drives how fast the robot starts speaking. Throughput / tokens-per-second (HIGHER better): how quickly the rest of the answer is generated. Nile-Chat-4B and Qwen3-4B lead both; Qwen3-8B and ALLaM-7B are slower.

Quality-speed tradeoff & VRAM

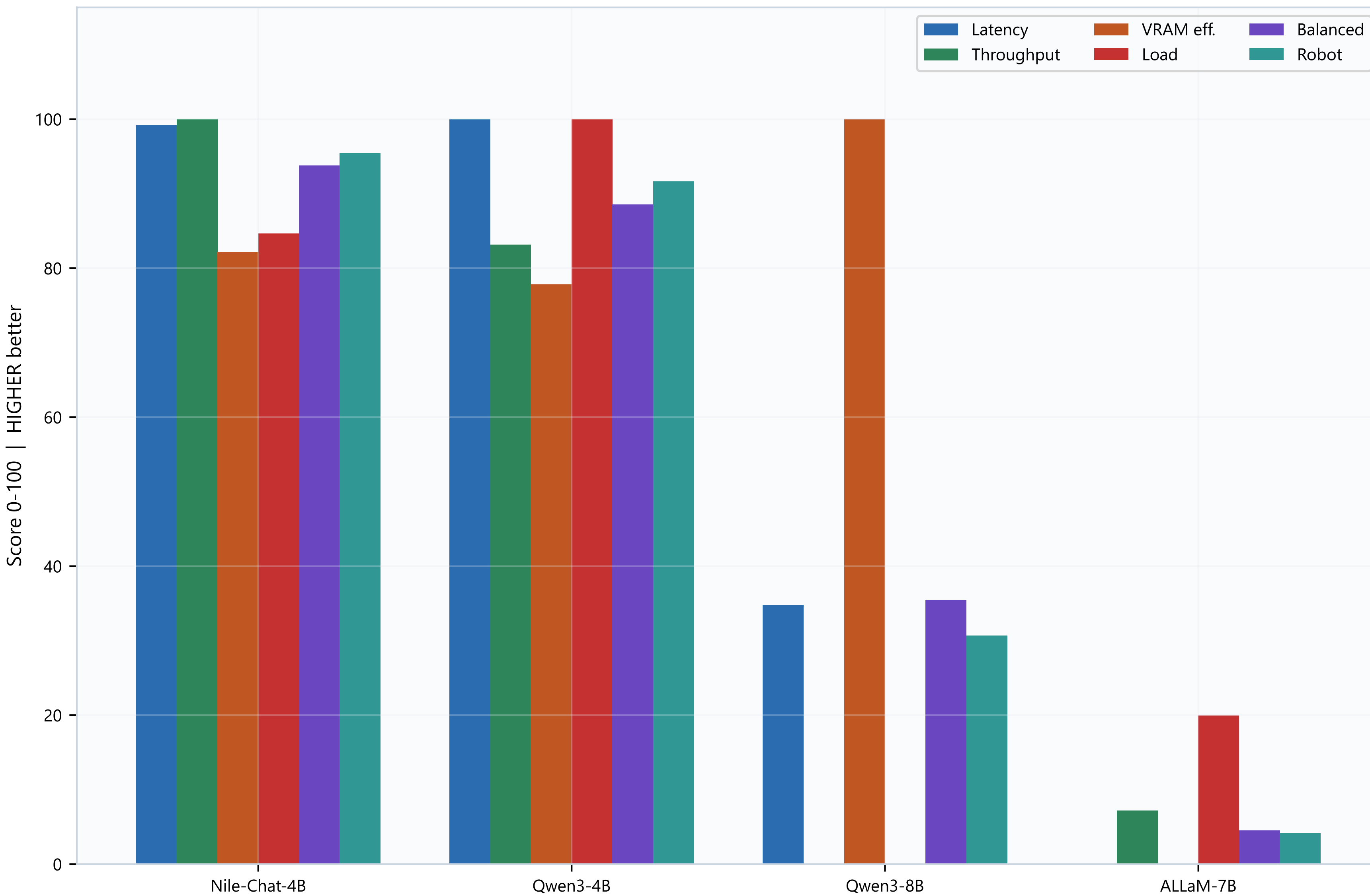




Tradeoff reading

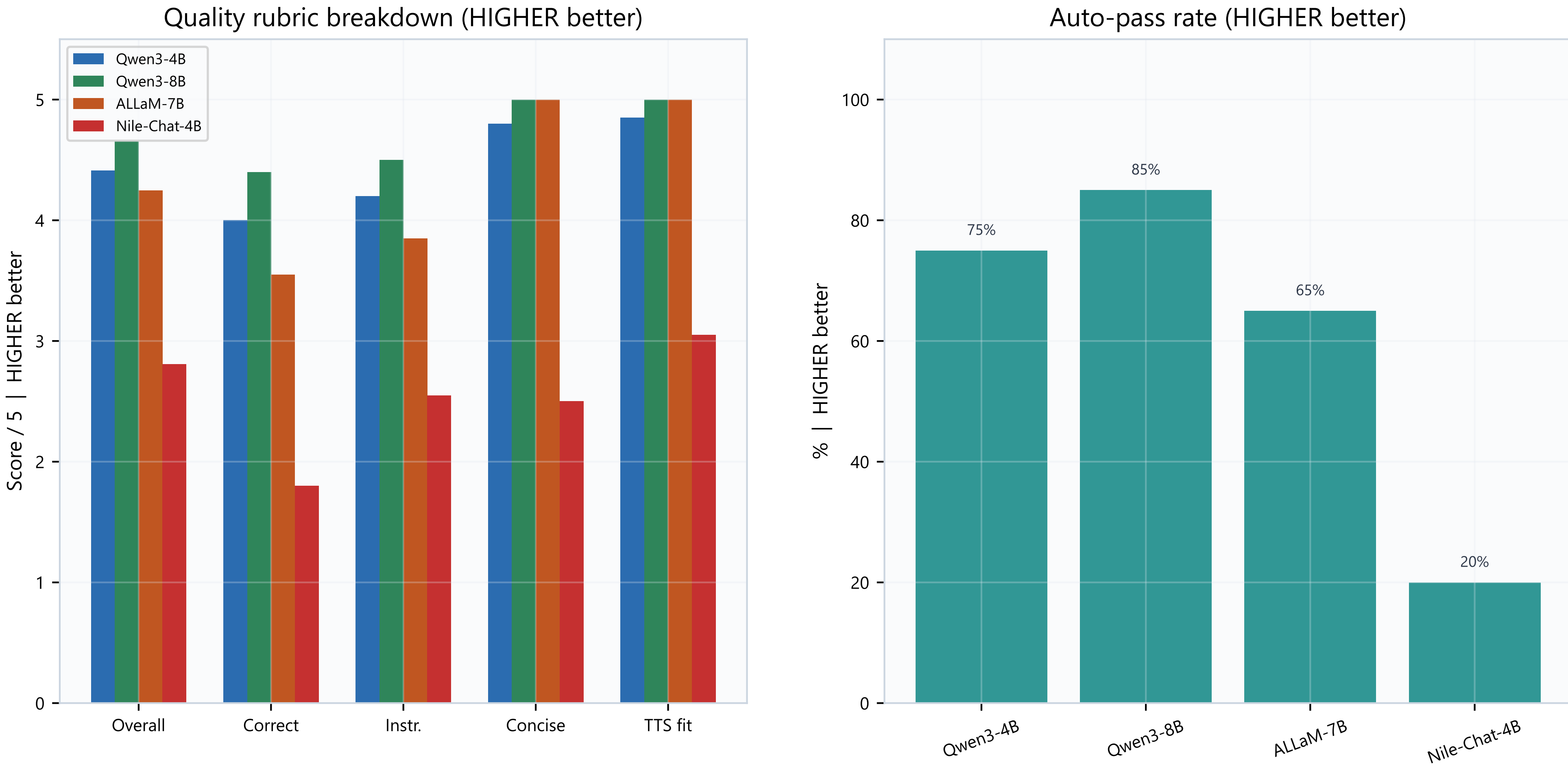
Bubble size = peak VRAM (LOWER better for co-residency with ASR/TTS).
Top-left of scatter is ideal: LOW TTFT + HIGH quality.
Qwen3-8B: higher quality, higher TTFT (slower), lowest VRAM (int4) — good if GPU is tight.
ALLaM-7B: near T4 ceiling (~15 GB) — risky when ASR+TTS share the GPU (VRAM LOWER better).
Nile-Chat-4B: fast TTFT but lower quality — often verbose (hurts TTS suitability).
Leave GPU headroom if ASR + LLM + TTS must share one card.

Normalized score breakdown (all scores: HIGHER better)



These are min-max normalized scores (HIGHER better on every bar). Latency score rewards LOW TTFT; Throughput rewards HIGH tok/s; VRAM efficiency rewards LOW peak VRAM; Load rewards LOW cold-load time. Robot score blends them for conversational robots.

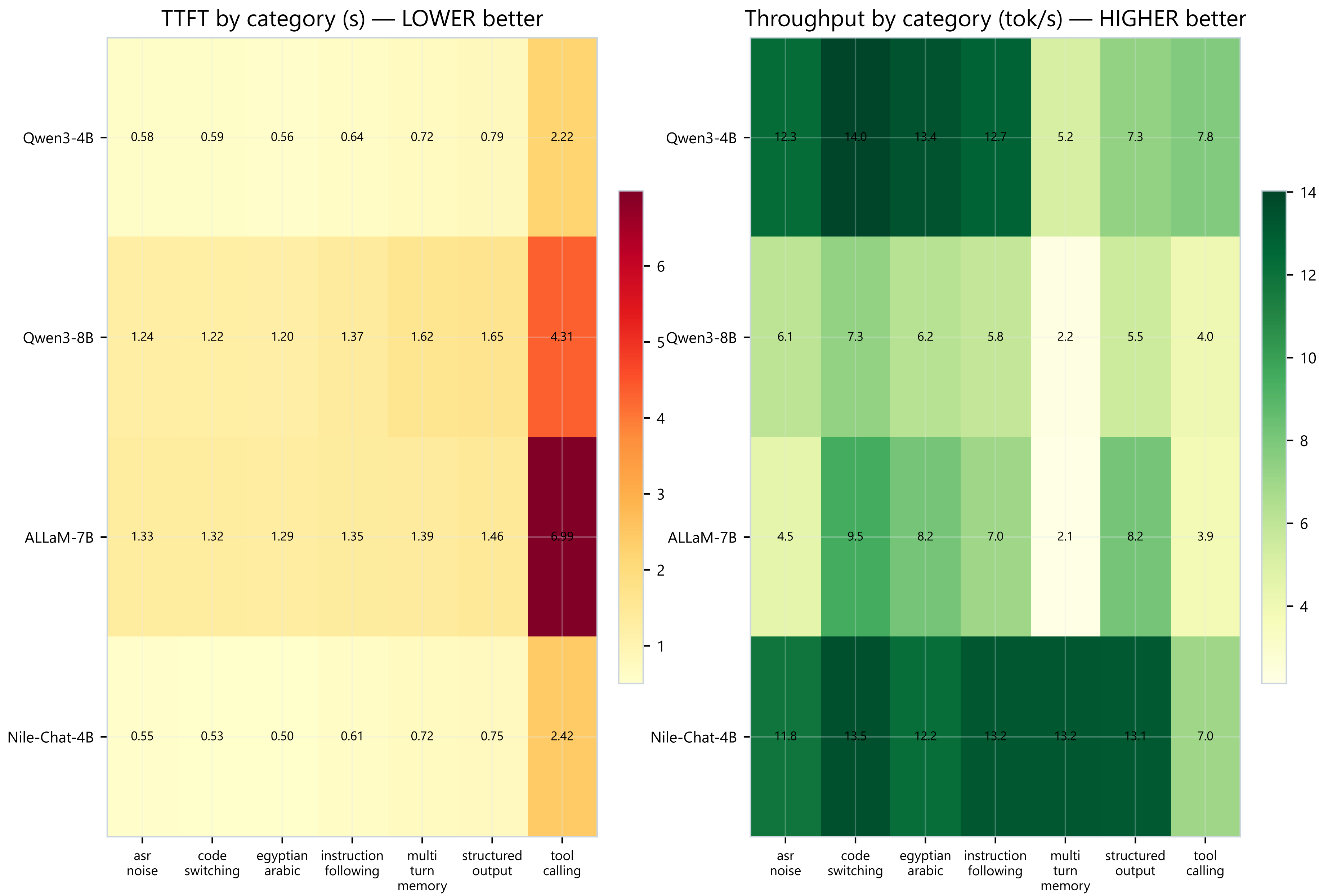
Quality dimensions & auto-pass rate (HIGHER better)



Quality interpretation

All quality metrics are HIGHER better (0-5 scale unless noted).
Correctness + instruction following matter most for robot reliability.
Conciseness and TTS suitability matter because spoken replies must stay short and clean.
Qwen3-8B leads overall quality (4.65/5) and auto-pass rate.
Nile-Chat-4B often over-generates (lower conciseness / TTS fit) despite fast tokens.
Language score was uniformly high for OK models — dialect capability is not the differentiator here.

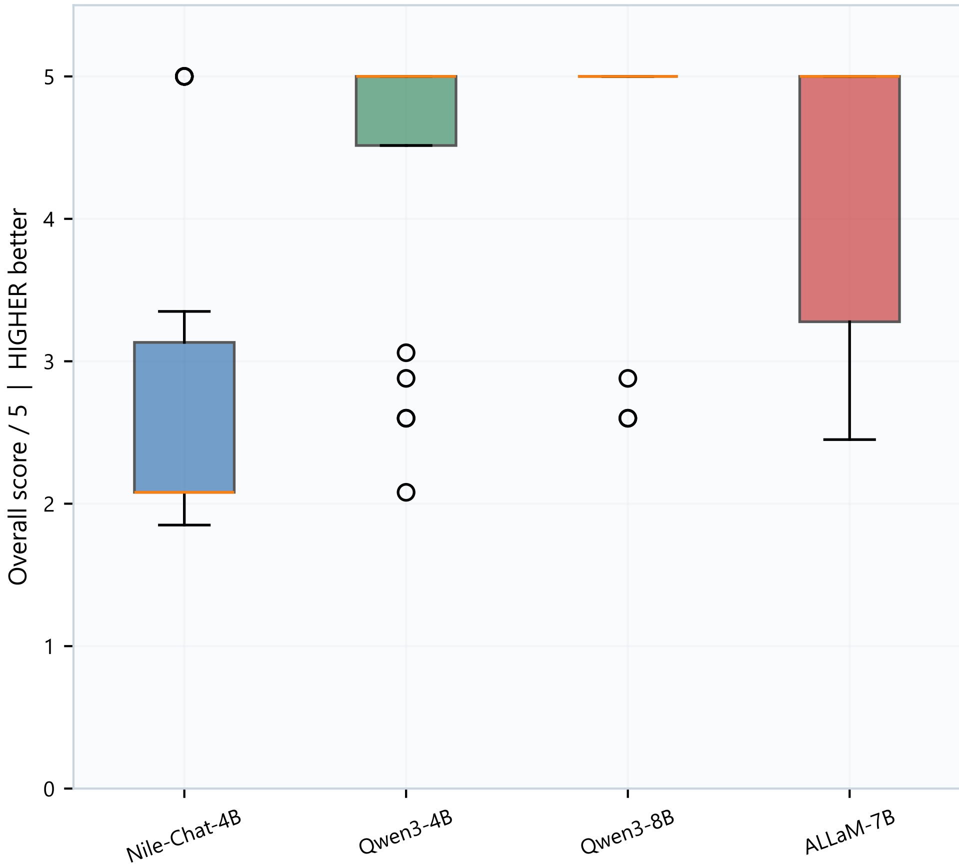
Category performance heatmaps



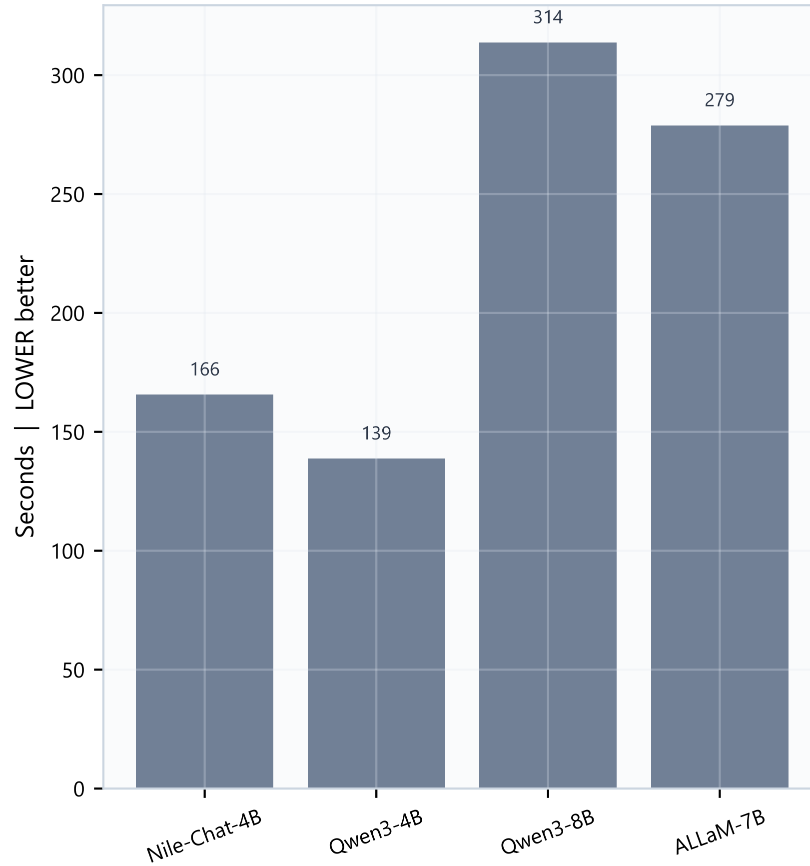
Left heatmap: TTFT seconds (LOWER better — darker usually means slower). Right heatmap: tokens/s (HIGHER better — darker green usually means faster generation). Tool-calling is the slowest category for all models.

Score distribution & decision guidance

Per-prompt quality spread (HIGHER better)



Cold load time (LOWER better)



How to choose

1) Need fastest conversational feel (LOW TTFT) -> Nile-Chat-4B.
2) Need best answer quality / auto-pass (HIGHER better) -> Qwen3-8B.
3) Need a balance on a mid GPU -> Qwen3-4B (strong TTFT + solid quality).
4) Need lowest VRAM (LOWER better) for co-residency -> Qwen3-8B (int4).
5) Avoid ALLaM-7B on T4-class GPUs unless dialect needs outweigh HIGH VRAM cost.
6) Gated models (Gemma / Jais) were not scored — re-run after HF auth if required.